STANDARD OPERATING PROCEDURE OF COMPRESSION SET VF-B10

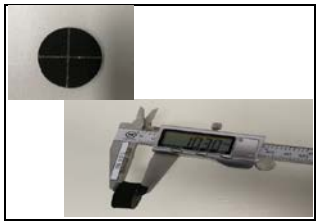
Lab /SOP/ 32



1. Compression platen
2. Spacers of thickness 4.0, 4.5, 5.0, 5.5, 6.0 mm.
3. Digital calipers capable of measuring up to 0.01mm
4. Air-circulating oven, capable of maintaining 45.0 ± 2 o C



Die cut four test samples having a diameter of 2.54 cm and a thickness of 10.0 ± 2.0 mm.

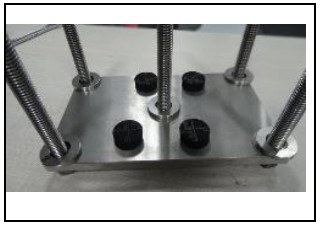


Draw 2 lines – could be parallel or across to each other.
Mark each line – ensure that the mark of each line is unique from the other line.
Measure the thickness of the sample, thus each specimen has 2 thickness readings and 2 results of compression set.



Place the appropriate spacers on the compression platen, that 50 % compression will be obtained.

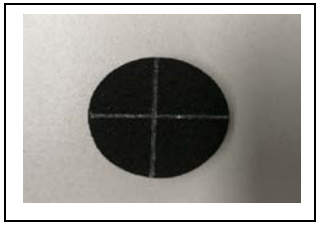
Sample thickness	Spacer thickness
8.0-8.5 mm	4.0 mm
8.5-9.5 mm	4.5 mm
9.5-10.5 mm	5.0 mm
10.5-11.5 mm	5.5 mm
11.5-12.0 mm	6.0 mm



Place the test specimen onto the bottom platen and position the top platen onto the test specimen.
Tighten the nuts on the test equipment until the top platen firmly contacts each of the spaces.



Within 15minutes after compressing the samples, place the compression set equipment with test samples into their appropriate environment.
EVA, Latex, Polyethylene, Polyurethane = 45 ± 2o C for 6 hours
Polyurethane – soft type utilized as backers, for Tongue and counter padding = 23 ± 2 o C for 22 hours



Allow the materials to recover at 23+/-2 o C
EVA, Latex, Polyethylene, Polyurethane = 30 ± 2 minutes
Polyurethane - soft type utilized as backer for Tongue and counter padding = 60 ± 2 minutes



Immediately following the completion of the recovery period, measure the thickness of the samples.
% Compression=Original thickness – Final thickness * 100
Original thickness - Spacer thickness
The average of the four test samples to the nearest 0.1%.

Prepared by	Approved by	REVISION		DATE	